

Exam Date & Time: 02-Jun-2023 (02:30 PM - 05:30 PM)



MANIPAL ACADEMY OF HIGHER EDUCATION

MANIPAL INSTITUTE OF TECHNOLOGY - BENGALURU
FOURTH SEMESTER B.TECH. DEGREE EXAMINATIONS - MAY/JUNE 2023

Machine Learning in VLSI computer aided design [ECE-4310]

Marks: 50

Duration: 180 mins.

PART-A

Answer all the questions.

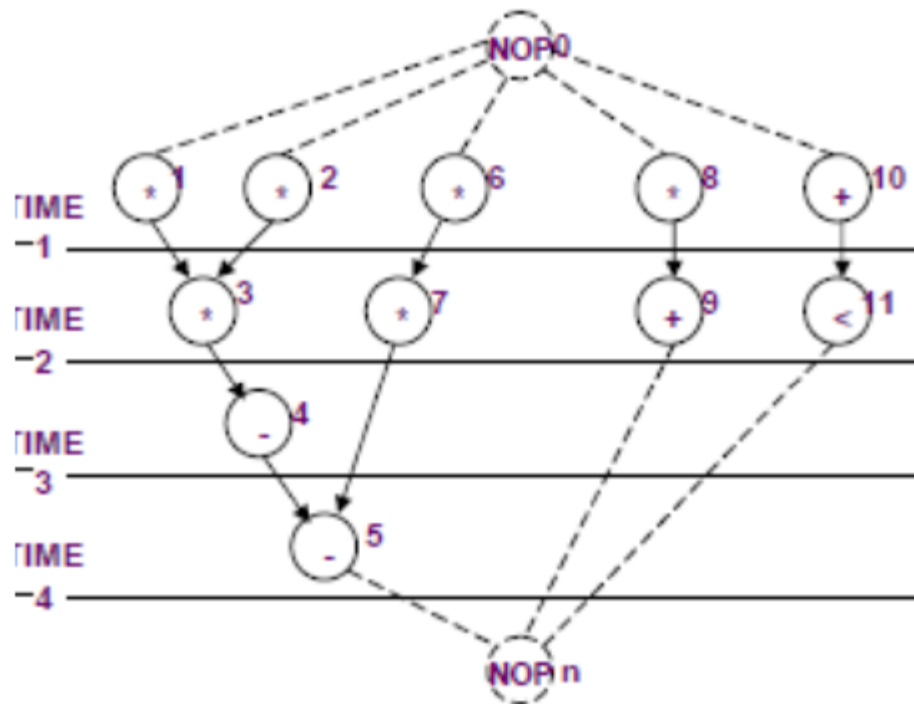
- 1) Implement $F = X_1'X_3'X_4' + X_1X_2 + X_3X_4$ using 4-2 input look-up table's and 1-3 input lookup table (4)
 - A)
 - B) Implement $y = abcd'e$ using ACT-2C module (3)
 - C) Explain VLSI design flow (3)
- 2) Draw ROBDD for 3-bit synchronous even counter. Show all the steps starting from OBDD. Perform ITE algorithm for ROBDD. (4)
 - A)
 - B) Draw ROBDD for 2 input XOR gate starting from BDD. Show all the steps. (3)
 - C) Draw Control and data flow graph for the following sequential statements $X = A * B$, $Y = C + D$, if $X < Y$ then $Z = Y - X$ else $Z = X - Y$. (3)
- 3) Using Quine Mccluskey method, obtain the minimized expression for the following function (4)

$$F = \sum m(0, 1, 6, 7, 8, 9, 13, 14, 15)$$
 - A)
 - B) Determine the prime implicants for the following function using iterated consensus method (3)

$$F = \sum m(0, 2, 3, 4, 5, 6)$$
 - C) Define the term exact minimization and heuristic minimization with an example each. (3)
- 4) Draw the sequencing graph for the given function $F = [(C + (A * B)) - ((B * C) + A)] < [P + Q] + Y$. Assume all the operations are having unit execution delay. Calculate latency using ASAP and mobility using ALAP algorithm (4)
 - A)
 - B) Explain ALAP algorithm with an example (3)
 - C) Define latency, mobility and resource constraints (3)

- 5) Apply Hu algorithm for the given data flow graph shown in the Figure. Assume $\gamma=4$, $P(0)=1, P(1)=3, P(2)=4, P(3)=2, P(4)=2$. Determine the resource constraints. Draw the scheduled graph with resource constraints

A)



(4)

B) Explain ESPRESSO algorithm with an example

(3)

C) Define RTL Schematic and technology schematic

(3)

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